

# Flask

Flask is a micro web framework written in Python.

# Install Flask

# Install Anaconda

The screenshot shows a web browser window with the URL `docs.anaconda.com/free/anaconda/install/index.html`. The page title is "Installation — Anaconda docu". The browser's address bar shows the URL and standard navigation icons. The page header includes the Anaconda logo, a navigation menu with links like "Free", "Starter", "Pro", "Business", "Enterprise", "Excel", "Reference", and "API", and a search icon. The main content area is titled "Installation" and includes a breadcrumb trail: `Home > Free tier > Anaconda Distribution > Installation`. The text on the page reads: "Review the system requirements listed below before installing Anaconda Distribution. If you don't want the hundreds of packages included with Anaconda, [install Miniconda](#), a mini version of Anaconda that includes just conda, its dependencies, and Python." A "Tip" box contains the text: "Looking for Python 3.5 or 3.6? See our [FAQ](#)." Below this is a "System requirements" section with a dropdown arrow. At the bottom of the page, there is a "We use cookies" notice with "Accept all" and "Reject all" buttons. A sidebar on the left titled "Section Navigation" lists various installation guides, with "Installation" currently selected.

Installation — Anaconda docu

docs.anaconda.com/free/anaconda/install/index.html

Free Starter Pro Business Enterprise Excel Reference API

Section Navigation

- Anaconda Distribution
  - Installation
    - Installing on Windows
    - Installing on macOS
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    - Installing on AWS Graviton2 (arm64)
    - Installing on Linux-s390x (IBM Z)
    - Installing on Linux POWER
    - Installing in silent mode
    - Installing for multiple users
    - Verifying your installation
    - Updating from older versions
    - Installing previous versions

Installation

Review the system requirements listed below before installing Anaconda Distribution. If you don't want the hundreds of packages included with Anaconda, [install Miniconda](#), a mini version of Anaconda that includes just conda, its dependencies, and Python.

**Tip**

Looking for Python 3.5 or 3.6? See our [FAQ](#).

**System requirements**

On Windows, macOS, and Linux, it is best to install Anaconda for the local user, which does not

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[More choices](#) [See our privacy policy](#)

# Create a virtual environment

- conda create -n myflask anaconda
- conda activate myflask

```
Anaconda Powershell Prompt (anaconda3)
(base) PS C:\Users\yjhuang> conda create -n myflask anaconda
Retrieving notices: ...working... done
Collecting package metadata (current_repodata.json): done
Solving environment: done

==> WARNING: A newer version of conda exists. <==
  current version: 23.1.0
  latest version: 23.10.0

Please update conda by running

  $ conda update -n base -c defaults conda

Or to minimize the number of packages updated during conda update use

  conda install conda=23.10.0

## Package Plan ##

  environment location: C:\Users\yjhuang\anaconda3\envs\myflask

  added / updated specs:
    - anaconda

The following packages will be downloaded:
```

| package           | build           | size   |
|-------------------|-----------------|--------|
| aiobotocore-2.5.0 | py311haa95532_0 | 151 KB |

```
webencodings pkgs/main/win-64::webencodings-0.5.1-py311haa95532_1
websocket-client pkgs/main/win-64::websocket-client-0.58.0-py311haa95532_4
werkzeug pkgs/main/win-64::werkzeug-2.2.3-py311haa95532_0
whatthepatch pkgs/main/win-64::whatthepatch-1.0.2-py311haa95532_0
wheel pkgs/main/win-64::wheel-0.38.4-py311haa95532_0
widgetsnbextension pkgs/main/win-64::widgetsnbextension-4.0.5-py311haa95532_0
win_inet_pton pkgs/main/win-64::win_inet_pton-1.1.0-py311haa95532_0
winpty pkgs/main/win-64::winpty-0.4.3-4
wrapt pkgs/main/win-64::wrapt-1.14.1-py311h2bbff1b_0
xarray pkgs/main/win-64::xarray-2023.6.0-py311haa95532_0
xlwings pkgs/main/win-64::xlwings-0.29.1-py311haa95532_0
xxhash pkgs/main/win-64::xxhash-0.8.0-h2bbff1b_3
xyzservices pkgs/main/win-64::xyzservices-2022.9.0-py311haa95532_1
xz pkgs/main/win-64::xz-5.4.2-h8cc25b3_0
y-py pkgs/main/win-64::y-py-0.5.9-py311hb6bf4ef_0
yaml pkgs/main/win-64::yaml-0.2.5-he774522_0
yapf pkgs/main/noarch::yapf-0.31.0-pyhd3eb1b0_0
yarl pkgs/main/win-64::yarl-1.8.1-py311h2bbff1b_0
ypy-websocket pkgs/main/win-64::ypy-websocket-0.8.2-py311haa95532_0
zeromq pkgs/main/win-64::zeromq-4.3.4-hd77b12b_0
zfp pkgs/main/win-64::zfp-1.0.0-hd77b12b_0
zict pkgs/main/win-64::zict-2.2.0-py311haa95532_0
zipp pkgs/main/win-64::zipp-3.11.0-py311haa95532_0
zlib pkgs/main/win-64::zlib-1.2.13-h8cc25b3_0
zlib-ng pkgs/main/win-64::zlib-ng-2.0.7-h2bbff1b_0
zope pkgs/main/win-64::zope-1.0-py311haa95532_1
zope.interface pkgs/main/win-64::zope.interface-5.4.0-py311h2bbff1b_0
zstd pkgs/main/win-64::zstd-1.5.5-hd43e919_0

Proceed ([y]/n)? y
```

# Install Flask

- pip install Flask

```
Anaconda Powershell Prompt (anaconda3)

done
#
# To activate this environment, use
#
#     $ conda activate myflask
#
# To deactivate an active environment, use
#
#     $ conda deactivate

(base) PS C:\Users\yjhuang>
(base) PS C:\Users\yjhuang> conda activate myflask
(myflask) PS C:\Users\yjhuang> pip install Flask
Requirement already satisfied: Flask in c:\users\yjhuang\anaconda3\envs\myflask\lib\site-packages (2.2.2)
Requirement already satisfied: Werkzeug>=2.2.2 in c:\users\yjhuang\anaconda3\envs\myflask\lib\site-packages (from Flask) (2.2.3)
Requirement already satisfied: Jinja2>=3.0 in c:\users\yjhuang\anaconda3\envs\myflask\lib\site-packages (from Flask) (3.1.2)
Requirement already satisfied: itsdangerous>=2.0 in c:\users\yjhuang\anaconda3\envs\myflask\lib\site-packages (from Flask) (2.0.1)
Requirement already satisfied: click>=8.0 in c:\users\yjhuang\anaconda3\envs\myflask\lib\site-packages (from Flask) (8.0.4)
Requirement already satisfied: colorama in c:\users\yjhuang\anaconda3\envs\myflask\lib\site-packages (from click>=8.0->Flask) (0.4.6)
Requirement already satisfied: MarkupSafe>=2.0 in c:\users\yjhuang\anaconda3\envs\myflask\lib\site-packages (from Jinja2>=3.0->Flask) (2.1.1)
(myflask) PS C:\Users\yjhuang> █
```

# Ensure Flask is installed correctly

- pip show flask

```
(myflask) PS C:\Users\royal\anaconda3\myflaskfolder> pip show flask
Name: Flask
Version: 3.1.2
Summary: A simple framework for building complex web applications.
Home-page:
Author:
Author-email:
License-Expression: BSD-3-Clause
Location: C:\Users\royal\anaconda3\envs\myflask\Lib\site-packages
Requires: blinker, click, itsdangerous, jinja2, markupsafe, werkzeug
Required-by:
(myflask) PS C:\Users\royal\anaconda3\myflaskfolder> |
```

# Save the following code as hello.py

```
from flask import Flask
app = Flask(__name__)
@app.route("/")
def hello():
    return "Hello, World!"
```

# Go to the folder of hello.py

- Ex.
- `cd C:\Users\yjhuang\Desktop\myflaskfolder`

```
(myflask) PS C:\Users\yjhuang> cd C:\Users\yjhuang\Desktop\myflaskfolder  
(myflask) PS C:\Users\yjhuang\Desktop\myflaskfolder>
```



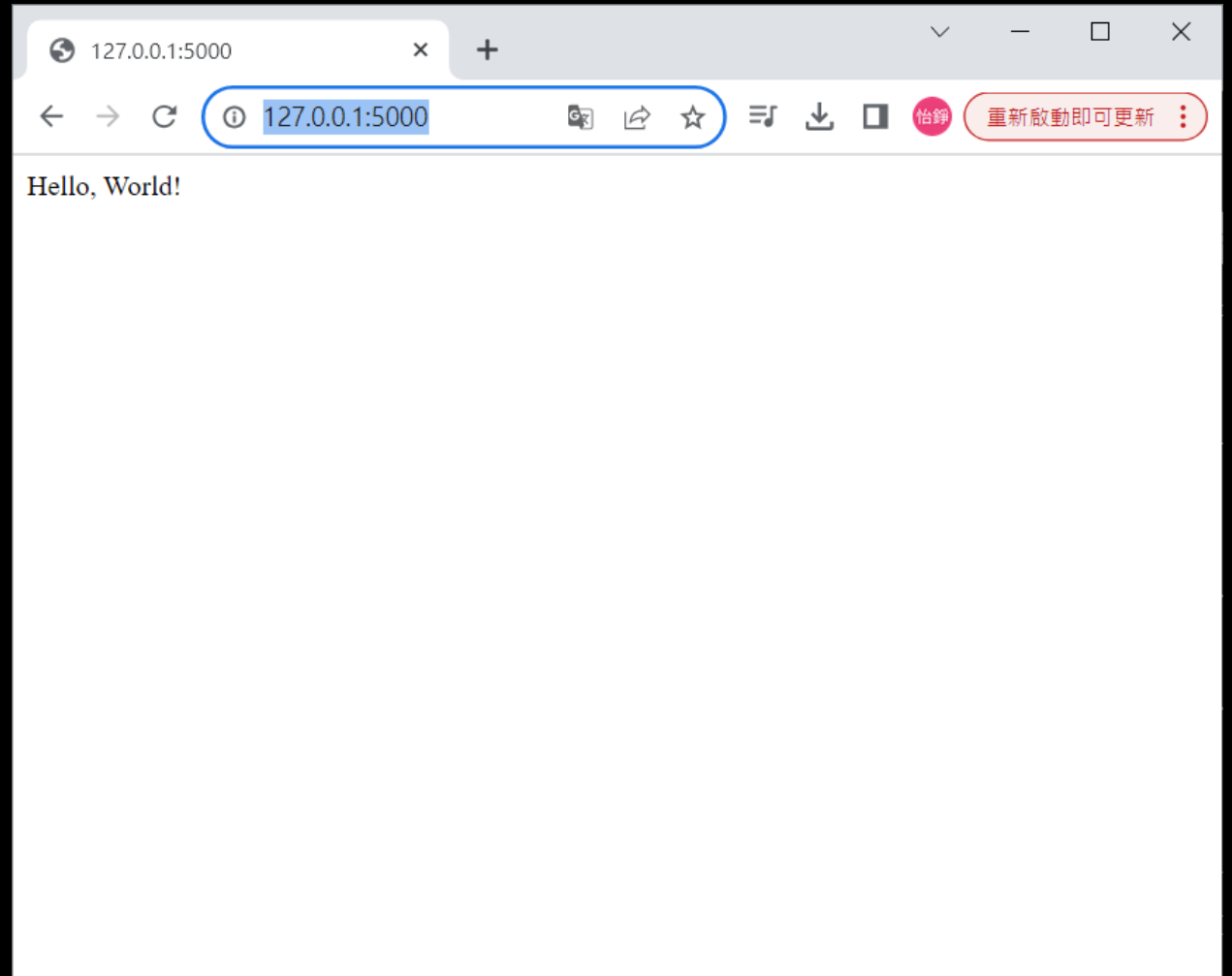
# Run

flask --app hello run

```
(myflask) PS C:\Users\yjhuang\Desktop\myflaskfolder> flask --app hello run
* Serving Flask app 'hello'
* Debug mode: off
WARNING: This is a development server. Do not use it in a production deployment
on WSGI server instead.
* Running on http://127.0.0.1:5000
Press CTRL+C to quit
127.0.0.1 - - [14/Nov/2023 11:59:30] "GET / HTTP/1.1" 200 -
127.0.0.1 - - [14/Nov/2023 11:59:30] "GET /favicon.ico HTTP/1.1" 404 -
```

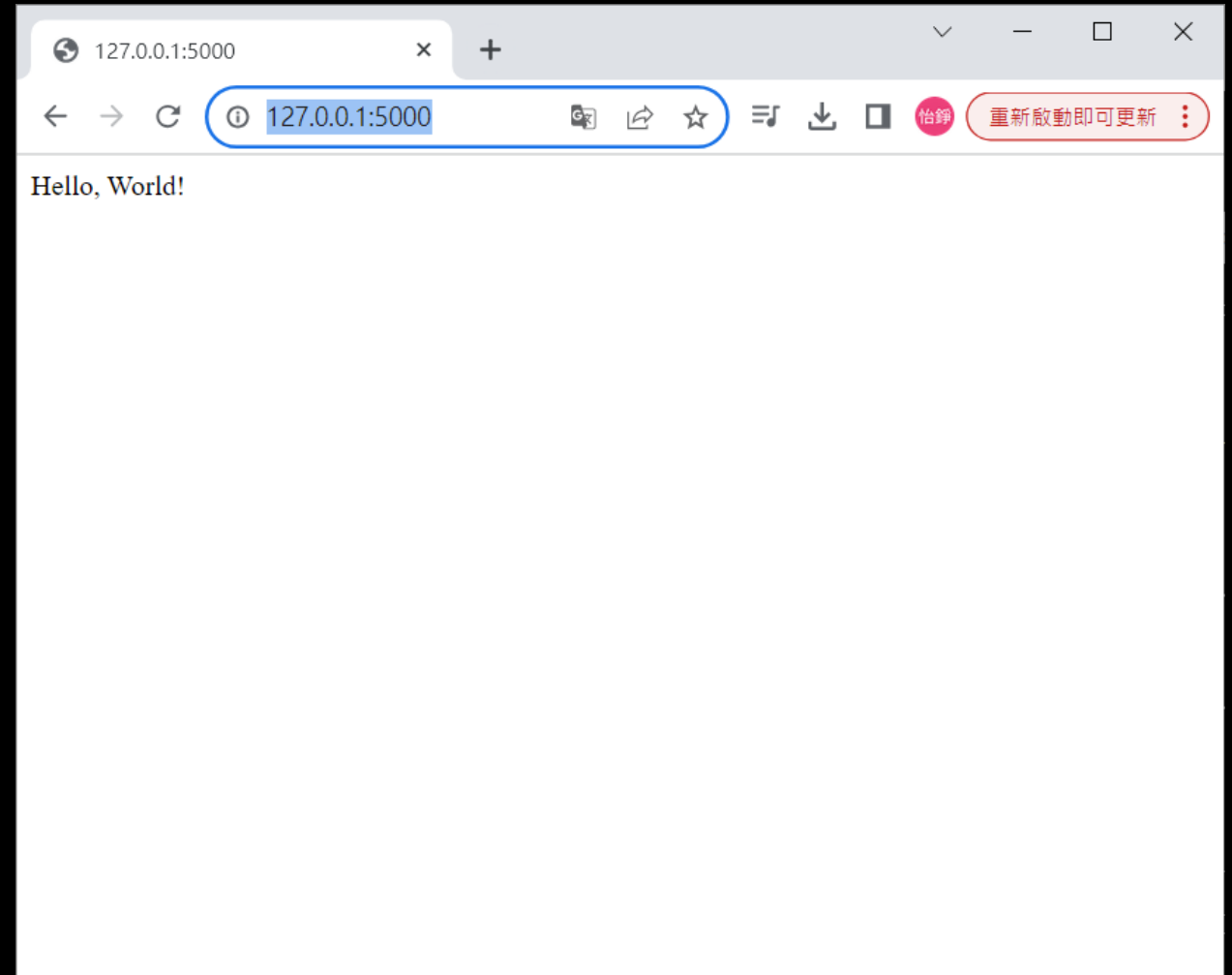
# Go to <http://127.0.0.1:5000/>

- See the Hello, World!



# Exercise 39: Hello Flask

- Create a local flask server
- Show that the server is running



# So what did that code do?

- First we imported the Flask class. An instance of this class will be our WSGI application.
- Next we create an instance of this class. The first argument is the name of the application's module or package. `__name__` is a convenient shortcut for this that is appropriate for most cases. This is needed so that Flask knows where to look for resources such as templates and static files.
- We then use the **`route()`** decorator to tell Flask what URL should trigger our function.
- The function returns the message we want to display in the user's browser. The default content type is HTML, so HTML in the string will be rendered by the browser.

# Routing

- Modern web applications use meaningful URLs to help users. Users are more likely to like a page and come back if the page uses a meaningful URL they can remember and use to directly visit a page.
- Use the route() decorator to bind a function to a URL.

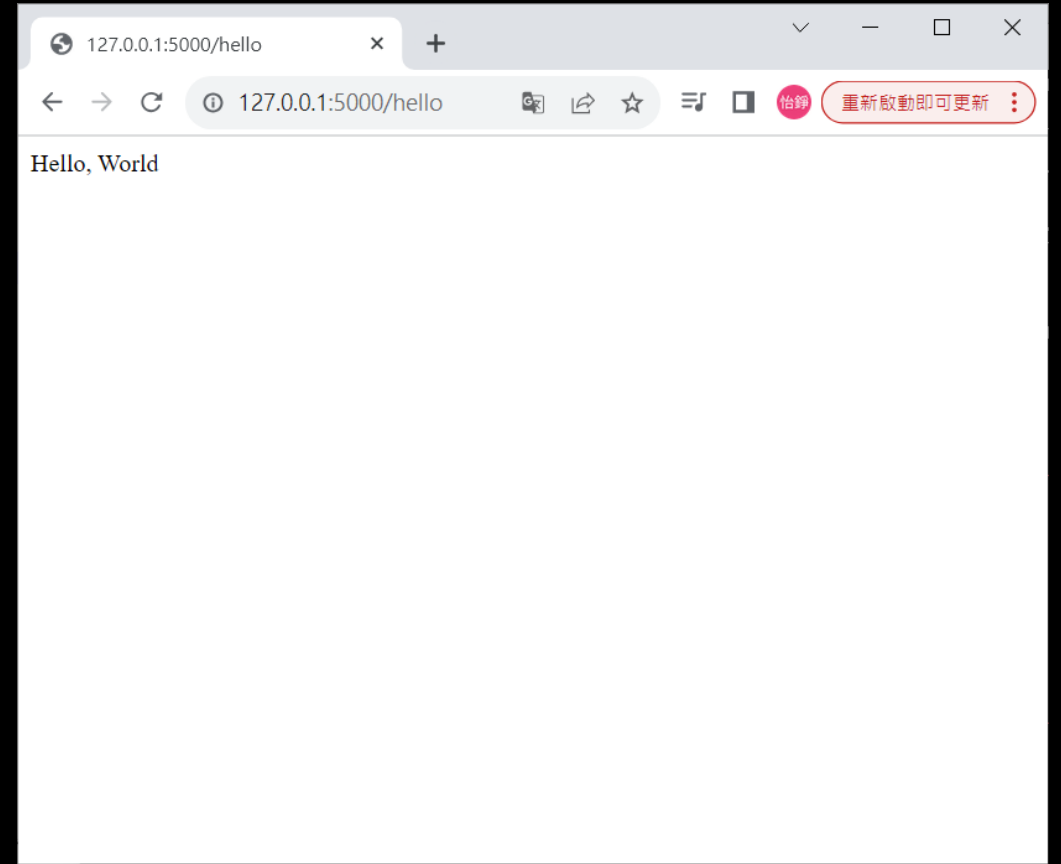
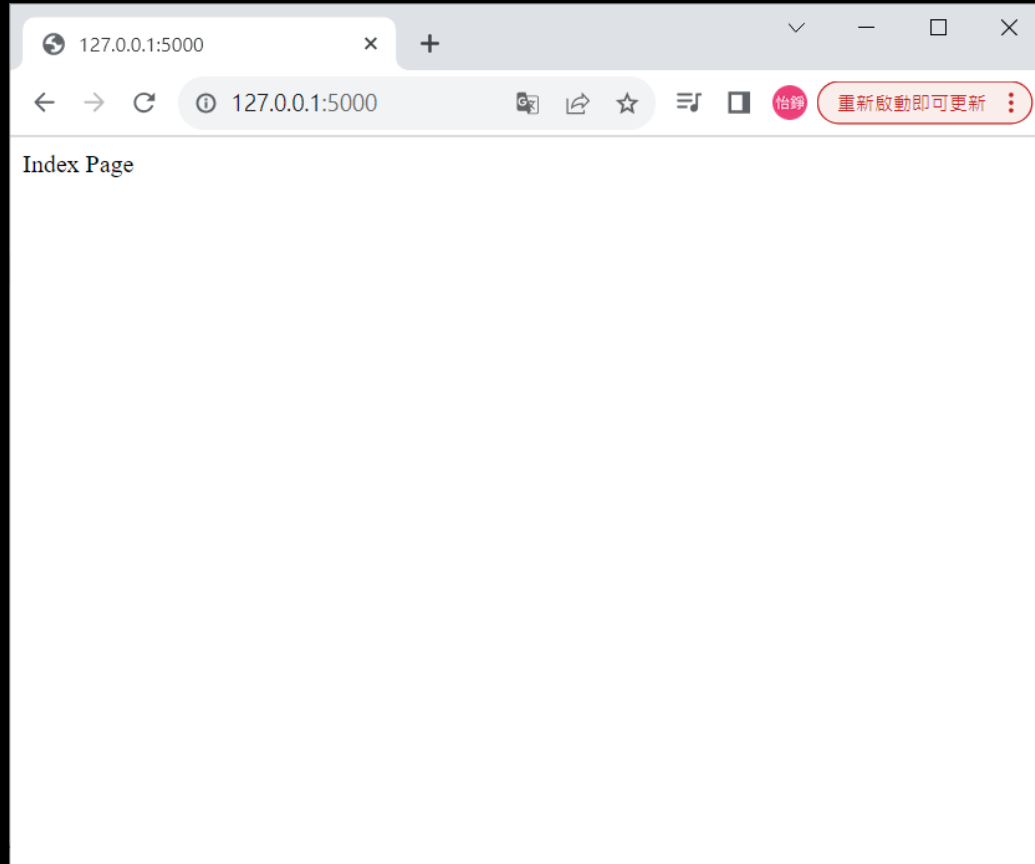
```
from flask import Flask

app = Flask(__name__)

@app.route('/')
def index():
    return 'Index Page'

@app.route('/hello')
def hello():
    return 'Hello, World'
```

# Different content for different address



# Variable Rules

- You can add variable sections to a URL by marking sections with `<variable_name>`. Your function then receives the `<variable_name>` as a keyword argument. Optionally, you can use a converter to specify the type of the argument like `<converter:variable_name>`.

|                     |                                                   |
|---------------------|---------------------------------------------------|
| <code>string</code> | (default) accepts any text without a slash        |
| <code>int</code>    | accepts positive integers                         |
| <code>float</code>  | accepts positive floating point values            |
| <code>path</code>   | like <code>string</code> but also accepts slashes |
| <code>uuid</code>   | accepts UUID strings                              |

```

from flask import Flask
from markupsafe import escape

app = Flask(__name__)

@app.route('/')
def index():
    return 'Index Page'

@app.route('/hello')
def hello():
    return 'Hello, World'

@app.route('/user/<username>')
def show_user_profile(username):
    # show the user profile for that user
    return f'User {escape(username)}'

@app.route('/post/<int:post_id>')
def show_post(post_id):
    # show the post with the given id, the id is an integer
    return f'Post {post_id}'

@app.route('/path/<path:subpath>')
def show_subpath(subpath):
    # show the subpath after /path/
    return f'Subpath {escape(subpath)}'

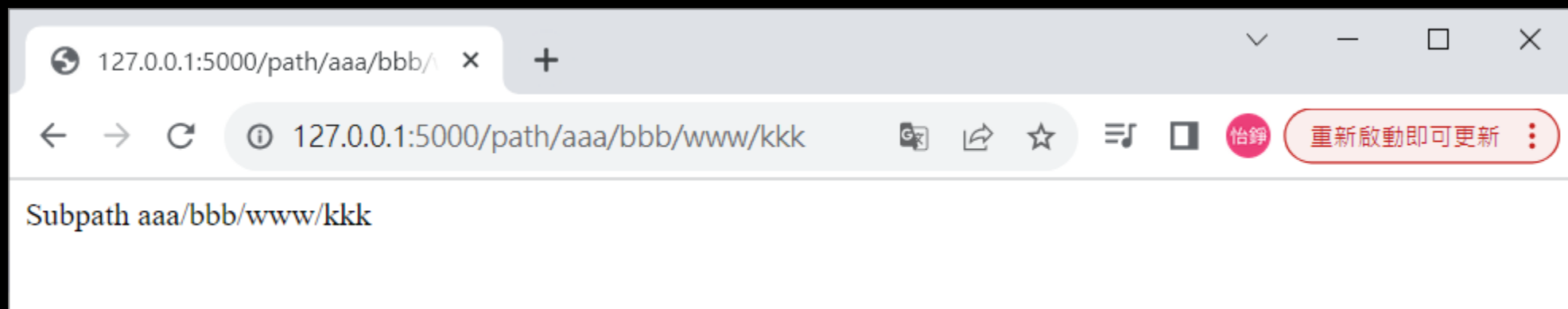
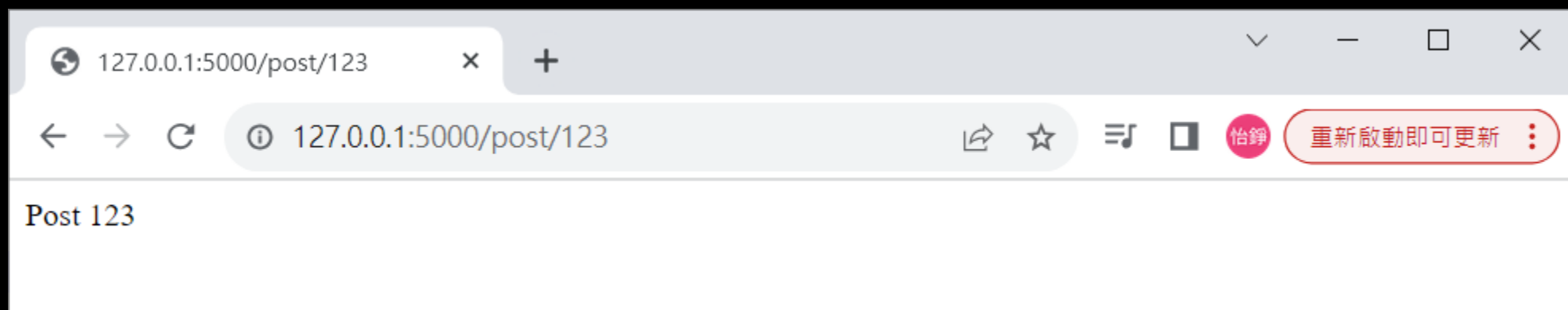
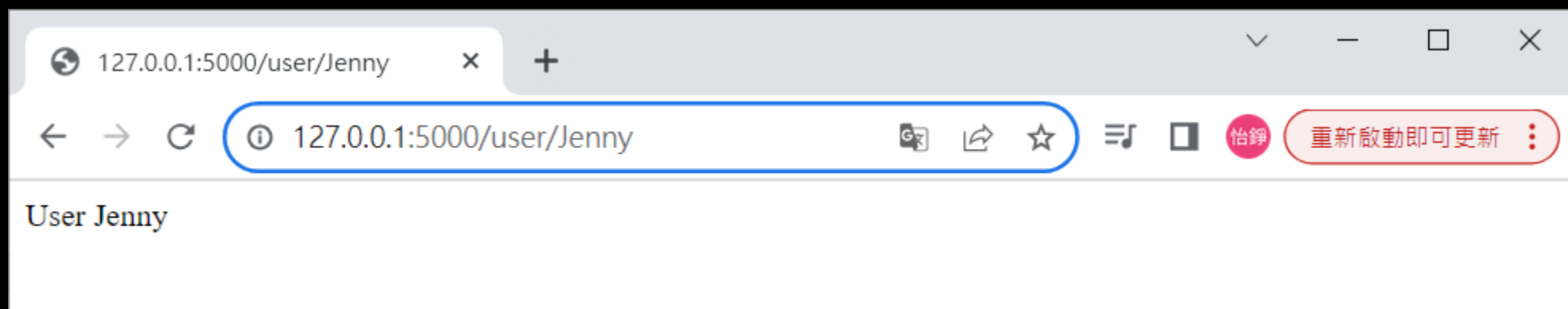
```

```

from flask import Flask
from markupsafe import escape
app = Flask(__name__)
@app.route('/')
def index():
    return 'Index Page'
@app.route('/hello')
def hello():
    return 'Hello, World'
@app.route('/user/<username>')
def show_user_profile(username):
    # show the user profile for that user
    return f'User {escape(username)}'
@app.route('/post/<int:post_id>')
def show_post(post_id):
    # show the post with the given id, the id is an integer
    return f'Post {post_id}'
@app.route('/path/<path:subpath>')
def show_subpath(subpath):
    # show the subpath after /path/
    return f'Subpath {escape(subpath)}'

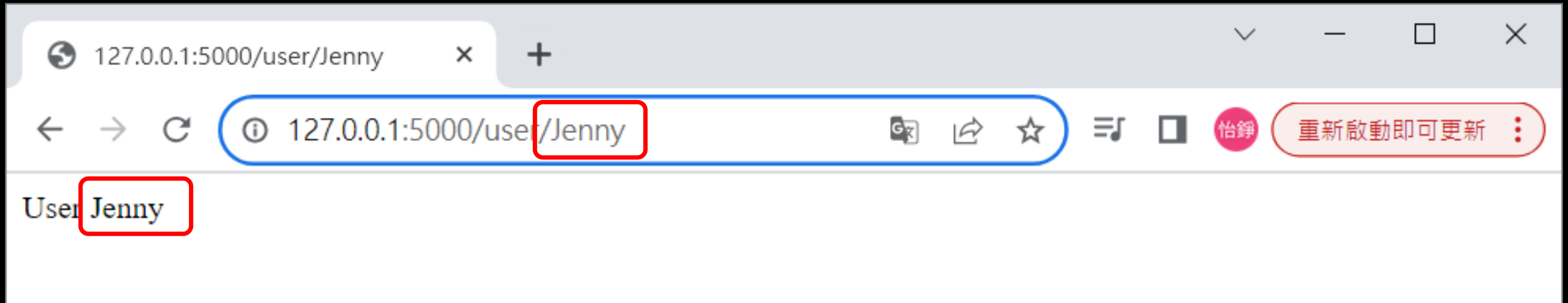
```





# Exercise 40: URL Info

- Show the information in the URL.
- For example:

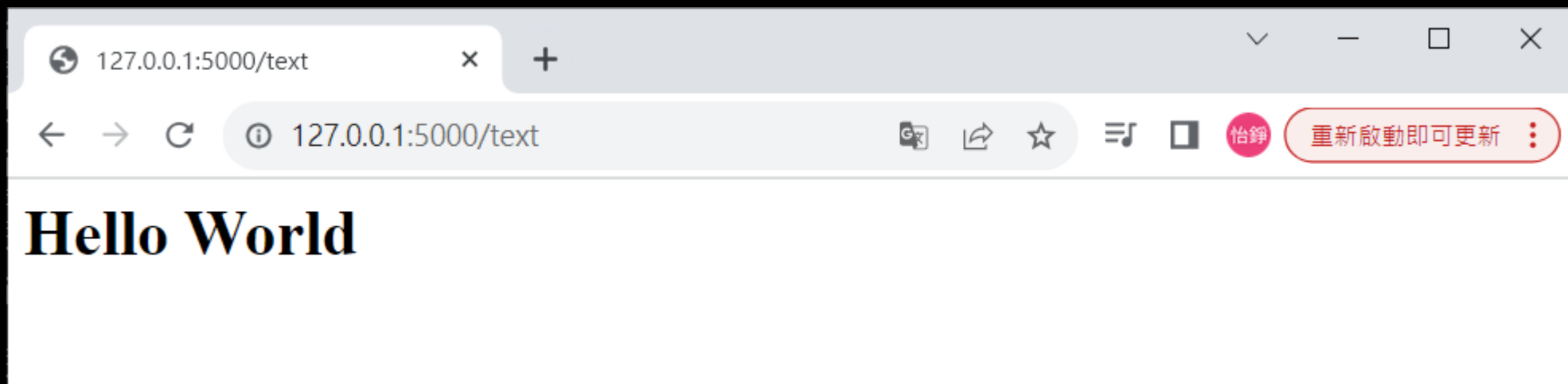


Simple HTML

```
from flask import Flask
from markupsafe import escape

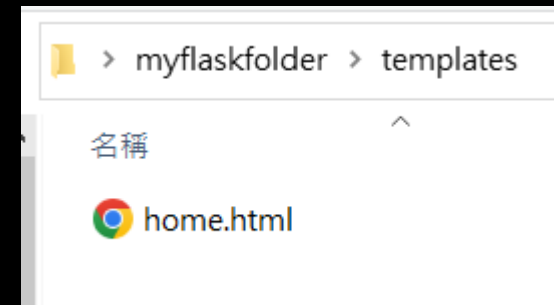
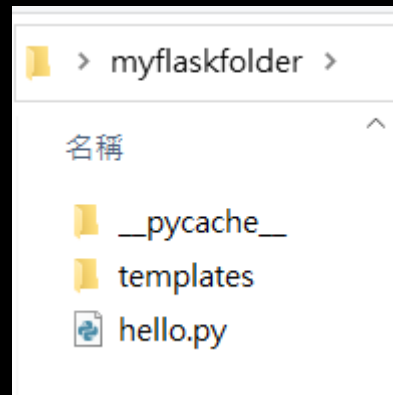
app = Flask(__name__)

@app.route('/text')
def text():
    return '<html><body><h1>Hello World</h1></body></html>'
```



Load HTML

- In the directory of the python executable, create a **templates** folder where the html file will be placed.
- /hello.py
- /templates
  - /home.html



# home.html

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <title>Home</title>
</head>
<body>
  <h1>My Website Text</h1>
  <table border="1">
    <tr>
      <td>Text Text Text</td>
      <td>Text Text Text</td>
      <td>Text Text Text</td>
    </tr>
    <tr>
      <td>Text Text Text</td>
      <td>Text Text Text</td>
      <td>Text Text Text</td>
    </tr>
    <tr>
      <td>Text Text Text</td>
      <td>Text Text Text</td>
      <td>Text Text Text</td>
    </tr>
  </table>
</body>
</html>
```

# hello.py

```
from flask import Flask, render_template
```

```
...
```

```
@app.route('/home')
```

```
def home():
```

```
    return render_template('home.html')
```

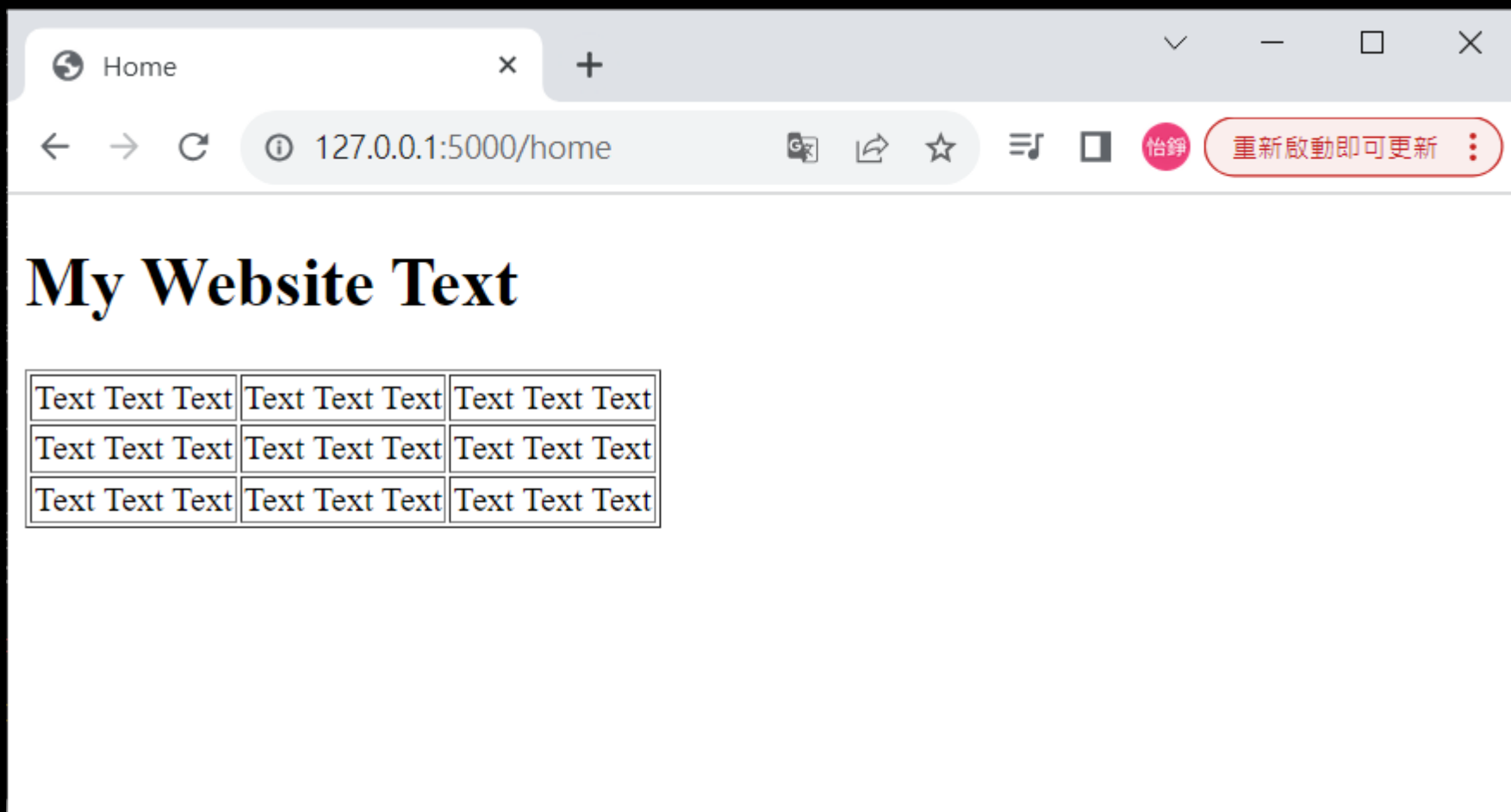
```
from flask import Flask, render_template
from markupsafe import escape

app = Flask(__name__)

@app.route('/home')
def home():
    return render_template('home.html')
```

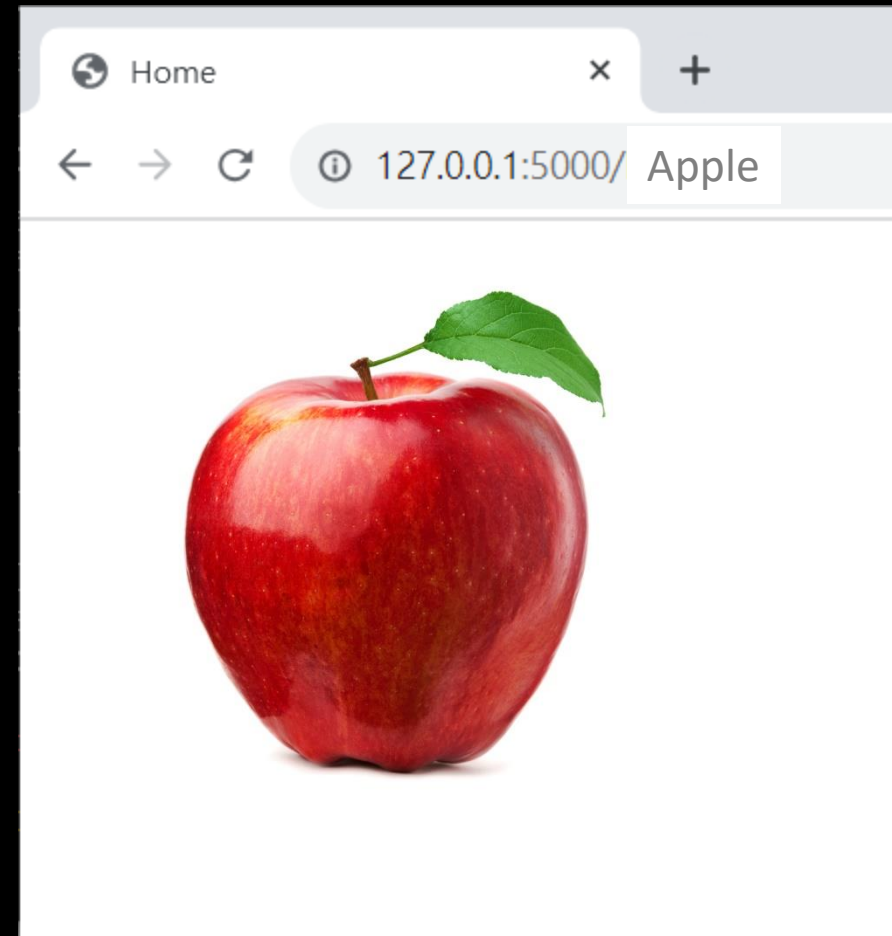
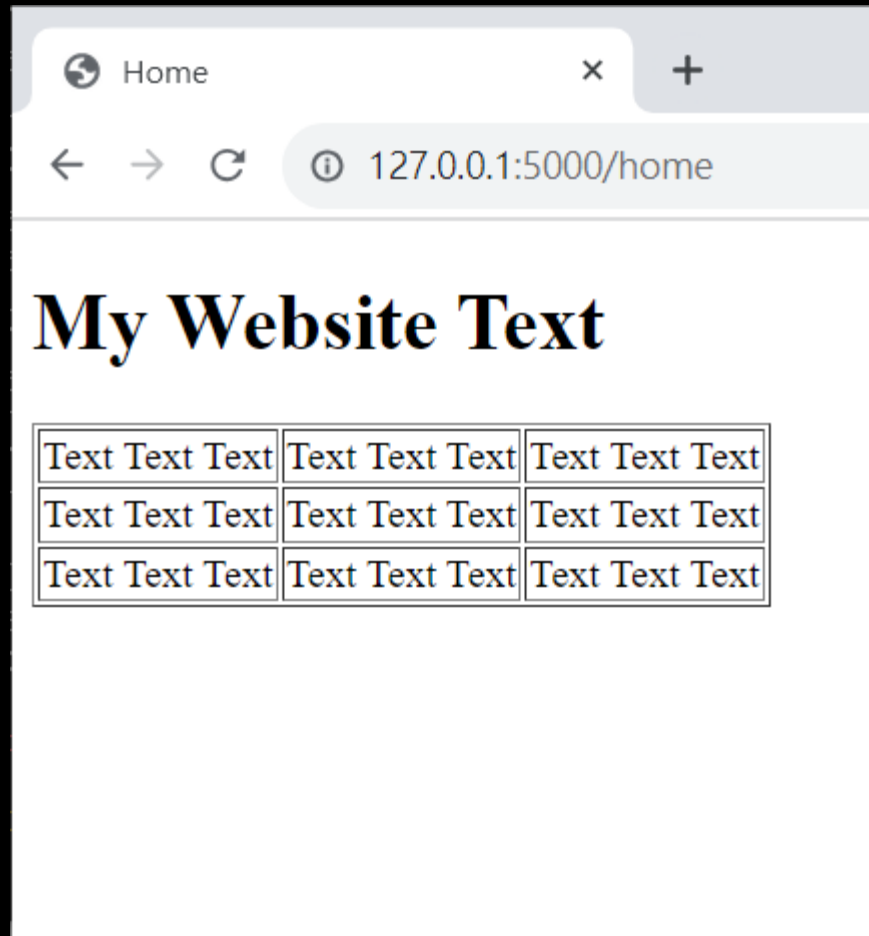


# Success!



# Exercise 41: Flask Load HTML

- Load different HTML for different URL
- For example:



Show Variables on HTML

- In the directory of the python executable, create a **templates** folder where the html file will be placed.
- /hello.py
- /templates
  - /page.html

# page.html

```
<!DOCTYPE html>
<html lang="en">

<head>
  <meta charset="UTF-8">
  <meta http-equiv="X-UA-Compatible" content="IE=edge">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Template - Page</title>
</head>

<body>
  <h1>Template - Page </h1>
  <h2>{{text}}</h2>

  {% if appInfo != undefined %}
  <h2>AppInfo : </h2>
  <p>id : {{appInfo.id}}</p>
  <p>name : {{appInfo.name}}</p>
  <p>version : {{appInfo.version}}</p>
  <p>author : {{appInfo.author}}</p>
  <p>remark : {{appInfo.remark}}</p>
  {% endif %}

</body>

</html>
```

```
<!DOCTYPE html>
<html lang="en">
```

```
<head>
  <meta charset="UTF-8">
  <meta http-equiv="X-UA-Compatible"
content="IE=edge">
  <meta name="viewport"
content="width=device-width, initial-
scale=1.0">
  <title>Template - Page</title>
</head>
```

```
<body>
  <h1>Template - Page </h1>
  <h2>{{text}}</h2>

  {% if appInfo != undefined %}
  <h2>AppInfo : </h2>
  <p>id : {{appInfo.id}}</p>
  <p>name : {{appInfo.name}}</p>
  <p>version : {{appInfo.version}}</p>
  <p>author : {{appInfo.author}}</p>
  <p>remark : {{appInfo.remark}}</p>
  {% endif %}
```

```
</body>
```

```
</html>
```

# hello.py

```
from flask import Flask, render_template
from markupsafe import escape
```

```
app = Flask(__name__)
```

```
@app.route('/page/app')
```

```
def pageAppInfo():
```

```
    appInfo = { # dict
```

```
        'id': 5,
```

```
        'name': 'Python - Flask',
```

```
        'version': '1.0.1',
```

```
        'author': 'Enoxs',
```

```
        'remark': 'Python - Web Framework'
```

```
    }
```

```
    return render_template('page.html', appInfo=appInfo, text="Python Flask !")
```

```
from flask import Flask, render_template
from markupsafe import escape

app = Flask(__name__)

@app.route('/page/app')
def pageAppInfo():
    appInfo = { # dict
        'id': 5,
        'name': 'Python - Flask',
        'version': '1.0.1',
        'author': 'Enoxs',
        'remark': 'Python - Web Framework'
    }
    return render_template('page.html', appInfo=appInfo, text="Python Flask !")
```

URL /info/

↓

Flask route

↓

pageAppInfo()

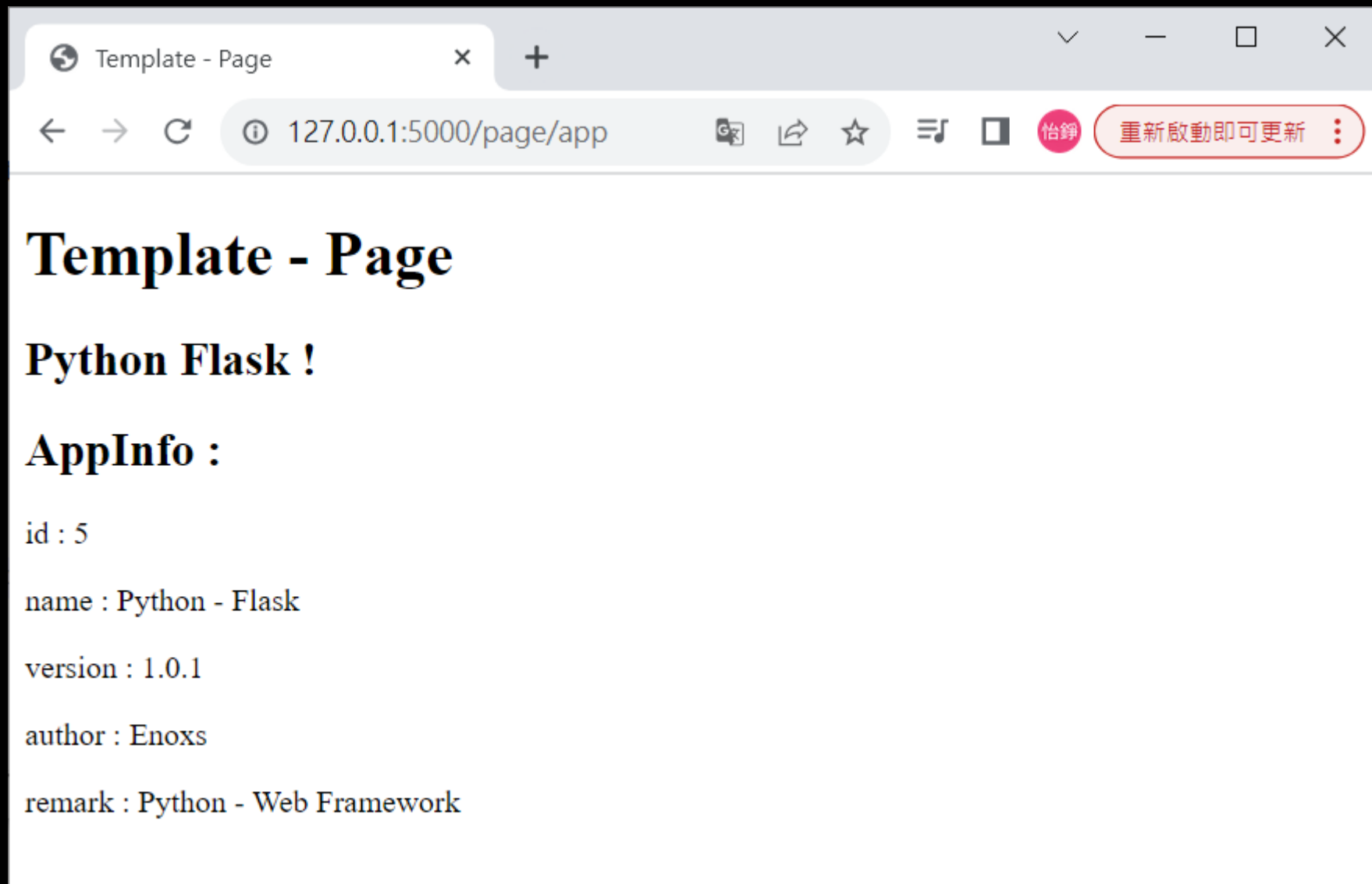
↓

render\_template("page.html", data)

↓

HTML renders on the browser ✓

# Success



# Exercise 42: Show Variables

- There is a Python object

```
x = {  
    "name": "John",  
    "age": 30,  
    "city": "New York"  
}
```

Show the object variables on HTML



# Deploy to Azure

<https://www.youtube.com/watch?v=foi3MiMcJYs>

Azure for Students: No payment method required. Please use your school email (@cc.ncu.edu.tw) for verification : [Azure | Microsoft Azure](#)

# Follow this:

- <https://learn.microsoft.com/en-us/azure/app-service/quickstart-python?tabs=flask%2Cwindows%2Cazure-cli%2Cvscode-deploy%2Cdeploy-instructions-azportal%2Cterminal-bash%2Cdeploy-instructions-zip-azcli>

## 2 - Create a web app in Azure

# Web App Configuration

The screenshot shows the Microsoft Azure portal interface. In the top navigation bar, the search bar contains 'web app'. The left sidebar shows the 'Azure 服務' (Azure Services) section with a plus icon and '快速入門中心' (Quickstart Center). The main content area displays search results for 'web app'. The 'Marketplace' tab is selected, showing a list of services. The 'Web 應用程式' (Web App) service is highlighted with a red box and a yellow callout that says 'Click this'. Below the 'Web 應用程式' service, there are links for '靜態 Web 應用程式' (Static Web App), 'Application Gateway', and 'Web 應用程式' (Web App). The '文件' (Docs) section shows links for 'Web 架構設計 - Azure Architecture Center' and '使用適合初學者的 ASP.NET Core · 建置 Web 應用程式 - Training'. The '瀏覽' (Browse) section shows a link for '繼續在 Microsoft Entra ID 中搜尋' (Continue searching in Microsoft Entra ID). The bottom of the page shows the '詢問 Copilot: 'web app'' (Ask Copilot: 'web app') section.

The screenshot shows the '建立 Web 應用程式' (Create Web App) wizard in the Azure portal. The wizard is divided into several sections: '基本' (Basic), '資料庫' (Database), '部署' (Deployment), '網路功能' (Networking), '監視器 + 安全' (Monitoring + Security), '標籤' (Tags), '檢閱 + 建立' (Review + Create). The '基本' section is currently active. It contains the following configuration steps:

- 訂用帳戶 \*** (Subscription): Azure for Students
- 資源群組 \*** (Resource Group): (新增) msdocs-python-webapp-quickstart (New)
- 執行個體詳細資料** (Instance Details):
  - 名稱** (Name): msdocs-python-webapp-quickstart-123
  - 安全唯一預設主機名稱開關** (Security Unique Default Host Name Switch): 開啟 (On)
- 發佈 \*** (Publish): 代碼 (Code)
- 執行階段堆疊 \*** (Runtime Stack): Python 3.13
- 作業系統 \*** (Operating System): Linux
- 地區 \*** (Region): Japan East
- 定價方案** (Pricing Plan): (新增) ASP-msdocspythonwebappquickstart-a5ed (New)
- 定價方案** (Pricing Plan): Premium V3 P0V3 (至少 195 的 ACU/vCPU, 4 GB 記憶體, 1 vCPU)

At the bottom, there are buttons for '檢閱 + 建立' (Review + Create), '< 上一步' (Previous Step), and '下一步: 資料庫 >' (Next Step: Database).

# 3 - Deploy your application code to Azure

# Web App Configuration (done)

首頁 > Microsoft.Web-WebApp-Portal-3167af58-b88e | 概觀

msdocs-python-webapp-quickstart-123 Web 應用程式

搜尋

瀏覽 停止 切換 重新啟動 刪除 重新整理 下載發行設定檔 重設發行設定檔 共用到行動裝置 傳送意見反應給我們

概觀

活動記錄

存取控制 (IAM)

標籤

診斷並解決問題

適用於雲端的 Microsoft Defender

事件 (預覽)

資源視覺化檢視

部署

設定

效能

App Service 方案

開發工具

API

監視

自動化

支援與疑難排解

程式集

資源群組 (移動) : [msdocs-python-webapp-quickstart](#)

狀態 : 正在執行

位置 (移動) : Japan East

訂用帳戶 (移動) : [Azure for Students](#)

訂用帳戶識別碼 : ad85b28f-114c-4e18-bdbc-2303b3ffab95

標籤 (編輯) : 新增標籤

預設網域 : [msdocs-python-webapp-quickstart-123-hkaqd6e0gae7dsfr.japaneast-01.azurewebsites.net](#)

App Service 方案 : [ASP-msdocspythonwebappquickstart-a5ed \(P0v3: 1\)](#)

作業系統 : Linux

健康情況檢查 : 未設定

JSON 檢視

屬性 正在監視 記錄 功能 通知 (0) 建議

Web 應用程式

名稱 : [msdocs-python-webapp-quickstart-123](#)

發佈模型 : 程式碼

執行階段堆疊 : Python - 3.13

執行階段狀態 : 良好

網域

預設網域 : [msdocs-python-webapp-quickstart-123...](#) 顯示較多

自訂網域 : [新增自訂網域](#)

裝載

方案類型 : App Service 方案

名稱 : [ASP-msdocspythonwebappquickstart-a5ed](#)

作業系統 : Linux

執行個體計數 : 1

SKU 和大小 : Premium0V3 (P0v3) 擴大

部署中心

部署記錄 : [檢視記錄](#)

上次部署 : 正在載入部署...

部署提供者 : None

Application Insights

名稱 : [啟用 Application Insights](#)

網路

虛擬 IP 位址 : 13.78.106.101

輸出 IP 位址 : 20.210.2.46, 20.210.6.26, 20.210.7.32, 2... 顯示較多

其他輸出 IP 位址 : 20.188.3.80, 40.115.161.55, 13.78.19.38,... 顯示較多

虛擬網路整合 : 未設定

按一下 Ctrl+Shift+F 以新增或移除我的最愛

# Deploy

Deploy center

msdocs-python-webapp-quickstart-123

Web 應用程式

搜尋

瀏覽 停止 切換 重新啟動 刪除 重新整理 下載發行設定檔 重設發行設定檔 共用到行動裝置 傳送意見反應給我們

概觀

活動記錄

存取控制 (IAM)

標籤

診斷並解決問題

適用於雲端的 Microsoft Defender

事件 (預覽)

資源視覺化檢視

部署

部署位置

部署中心

設定

環境變數

組態

執行個體

驗證

身分識別

備份

自訂網域

憑證

網路

Web 工作

服務連接器

程式集

資源群組 (移動) : msdocs-python-webapp-quickstart

狀態 : 正在執行

位置 (移動) : Japan East

訂閱帳戶 (移動) : Azure for Students

訂閱帳戶識別碼 : ad85b28f-114c-4e18-bdbc-2303b3ffab95

標籤 (編輯) : 新增標籤

JSON 檢視

預設網域 : msdocs-python-webapp-quickstart-123-hkaqd6e0gae7dsfr.japan...

App Service 方案 : ASP-msdocspythonwebappquickstart-a5ed (P0v3: 1)

作業系統 : Linux

健康情況檢查 : 未設定

屬性 正在監視 記錄 功能 通知 建議

Web 應用程式

名稱 : msdocs-python-webapp-quickstart-123

發佈模型 : 程式碼

執行階段堆疊 : Python - 3.13

執行階段狀態 : 良好

網路

預設網域 : msdocs-python-webapp-quickstart-123-hkaqd6e0gae7dsfr.japaneast-01.azurewebsites.net

自訂網域 : 新增自訂網域

裝載

方案類型 : App Service 方案

名稱 : ASP-msdocspythonwebappquickstart-a5ed

部署中心

部署記錄 : 檢視記錄

上次部署 : 正在載入部署...

部署提供者 : None

Application Insights

名稱 : 啟用 Application Insights

網路

虛擬 IP 位址 : 13.78.106.101

輸出 IP 位址 : 20.210.2.46, 20.210.6.26, 20.210.7.32, 20.210.7.34, 20.210.7.76, 20.210.7.160, 13.78.106.101

其他輸出 IP 位址 : 20.188.3.80, 40.115.161.55, 13.78.19.38, 52.140.240.157, 52.185.175.220, 52.140.231.76, 40.115.233.130, 13.73.20.60, 52.246.189.132, 20.210.2.46, 20.210.6.26, 20.210.7.32,

# Deploy

msdocs-python-webapp-quickstart-123 | 部署中心 ☆ ...

Web 應用程式

搜尋

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標籤

診斷並解決問題

適用於雲端的 Microsoft Defender

事件 (預覽)

資源視覺化檢視

部署

部署位置

部署中心

設定

設定 容器 (新增) 記錄 FTPS 認證

儲存 ✕ 捨棄 ↺ 重新整理 瀏覽 同步 傳送意見反應給我們

您現在處於生產位置，不建議設定 CI/CD。 [深入了解](#)

從您的常用的來源與組建提供者，部署及建立程式碼。 [深入了解](#)

來源 \* 發行檔案 (新)

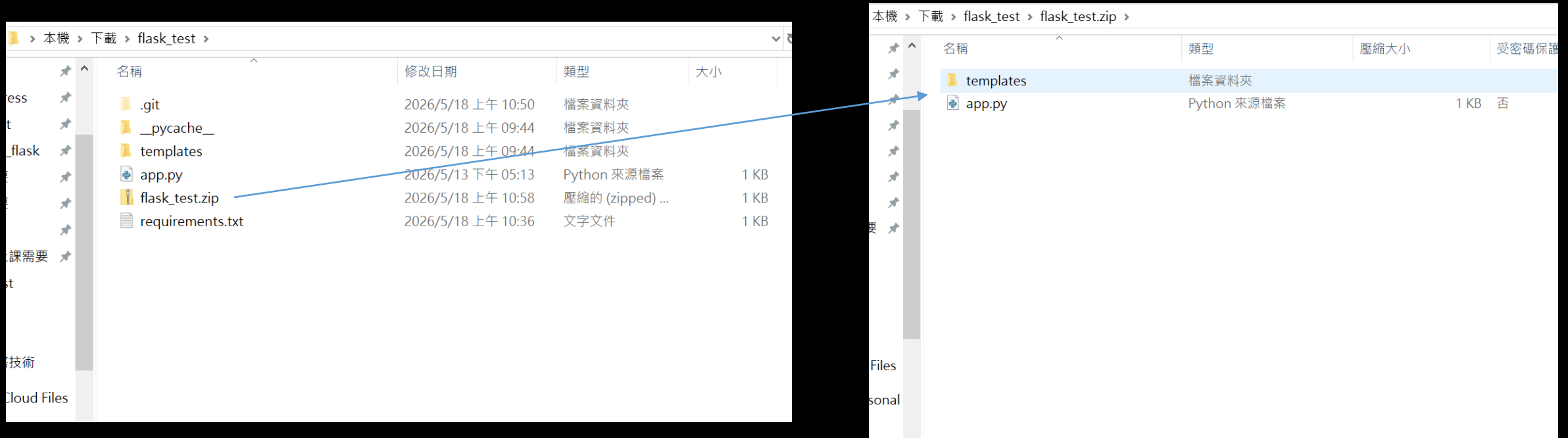
發行檔案

使用下列表單來選擇您的 ZIP 套件 (最大檔案大小 2 GB)。然後，按一下 [儲存] 以上傳。上傳完成後，網站將自動重新啟動。您可以在 [記錄] 區段中監控進度。 [深入了解](#)

flask\_test.zip 瀏覽



# Deployment file structure



Ensure that the project files are at the root of the ZIP archive; do not include an extra subfolder.

# Azure Deployment Demo

- Template - Page



# Shell-based

- 
- `$appServiceName='firstpage'`
- `$resourceGroupName='DefaultResourceGroup-EJP'`
- 
- `az webapp config appsettings set --name $appServiceName --resource-group $resourceGroupName --settings SCM_DO_BUILD_DURING_DEPLOYMENT=true`
- `az webapp deploy --name $appServiceName --resource-group $resourceGroupName --src-path double.zip --type zip`

When you deploy your app (for example from GitHub, VS Code, or ZIP):

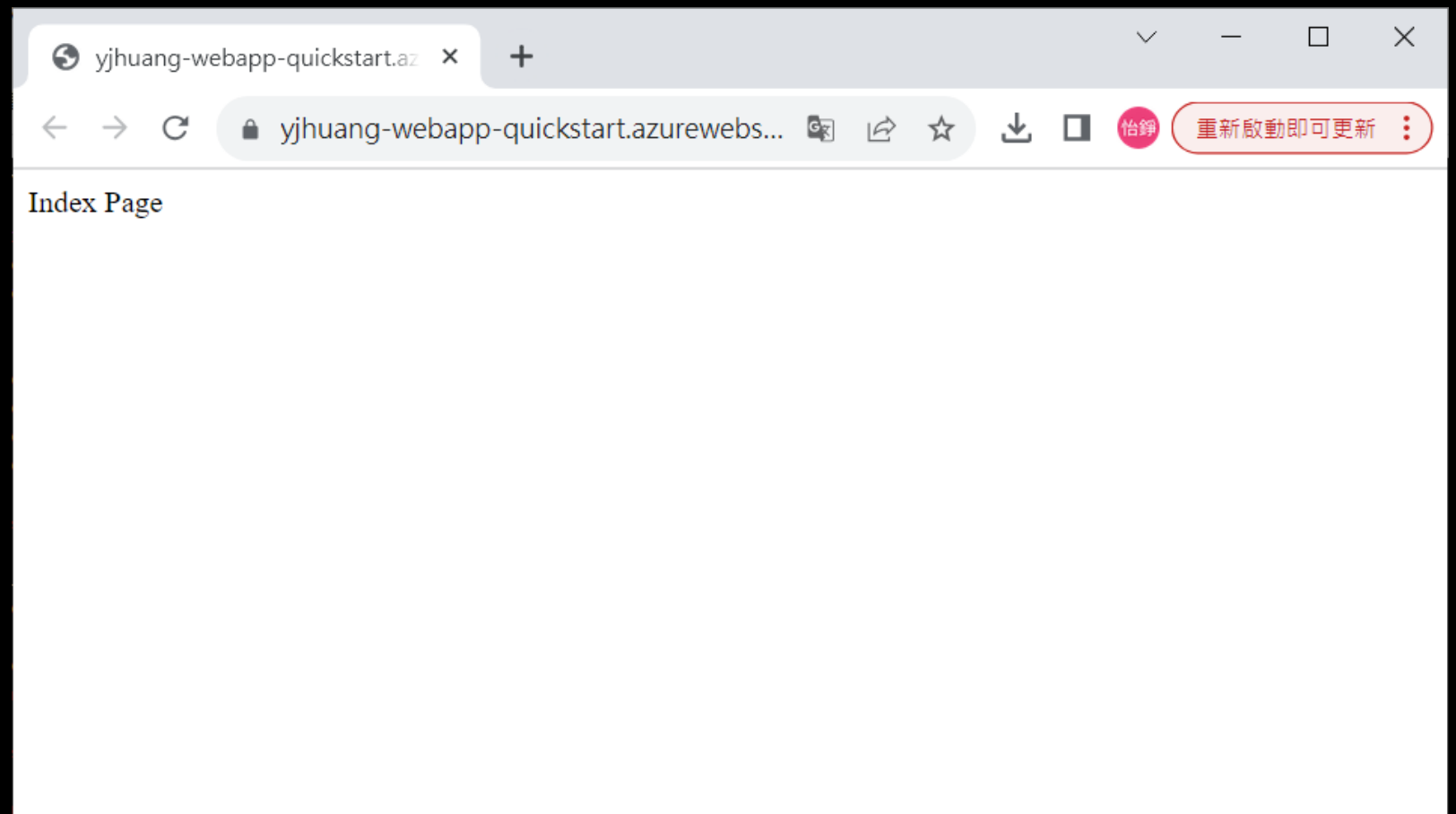
- If this setting is TRUE: 🖱️ Azure will automatically:
  - install dependencies (like `pip install -r requirements.txt`)
  - build the project
  - prepare it to run
- If FALSE (or not set): 🖱️ Azure will just copy files → no build, no dependency install

Substitute by yours



# Exercise 43: Flask Online

- Deploy your flask code online

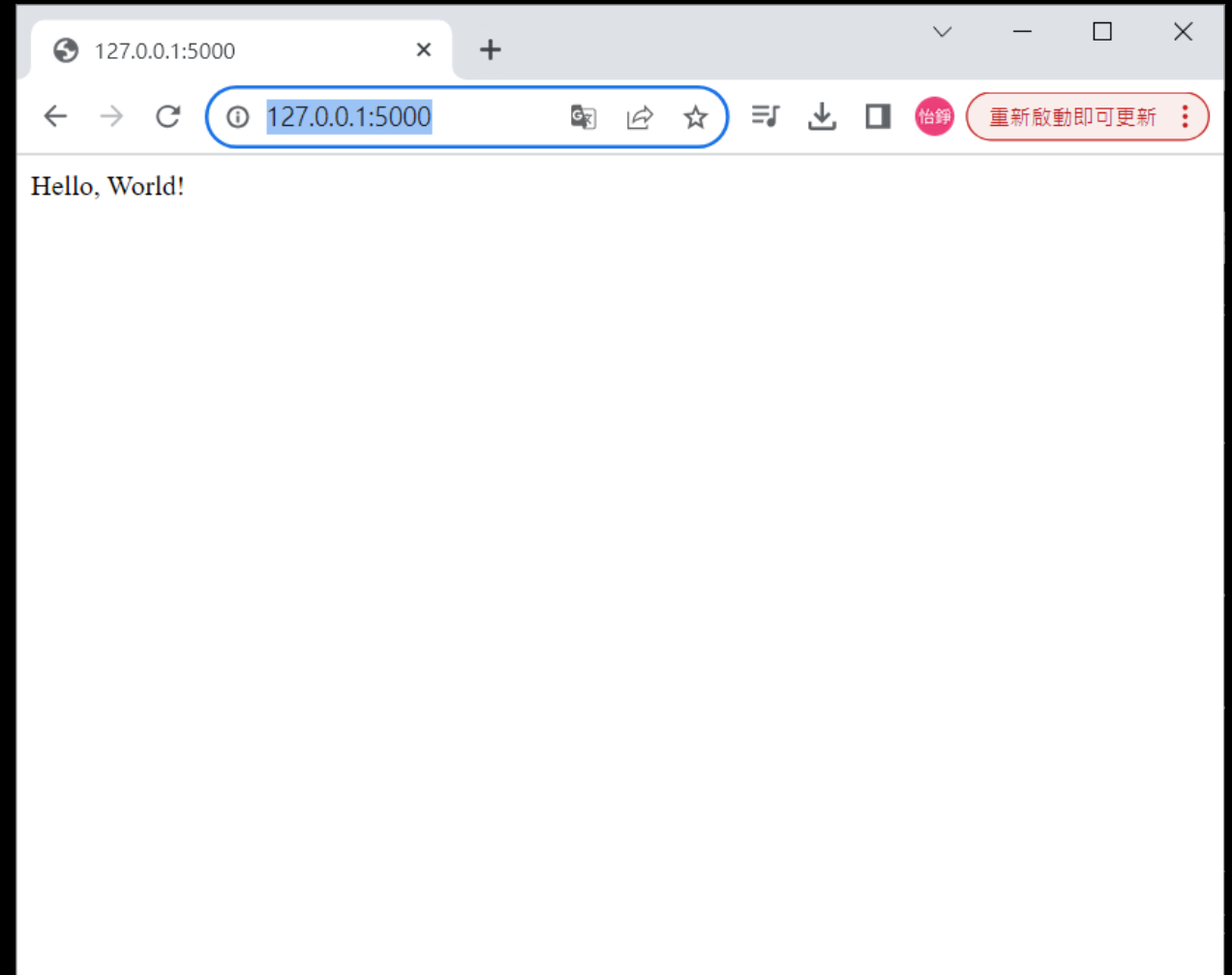


# More Flask Resource

- <https://flask.palletsprojects.com/en/3.0.x/>
- <https://ithelp.ithome.com.tw/articles/10258223>

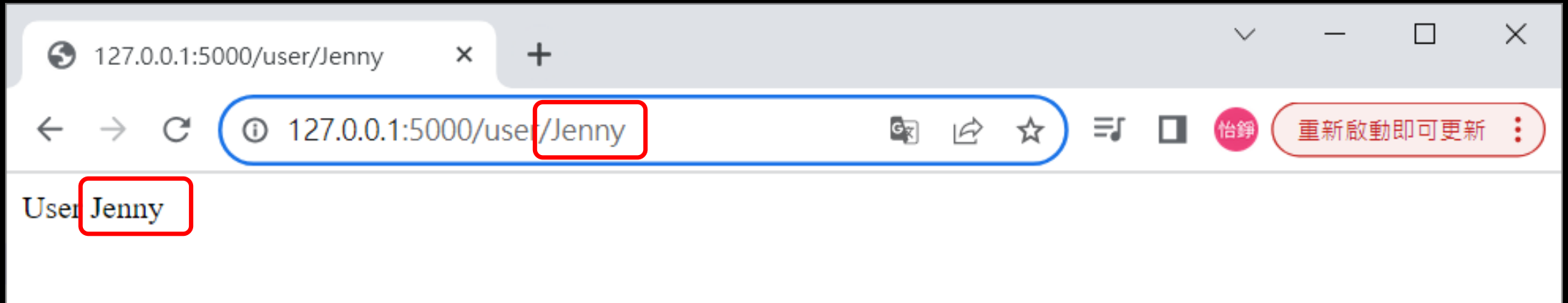
# Exercise 44: Hello Flask

- Create a local flask server
- Show that the server is running



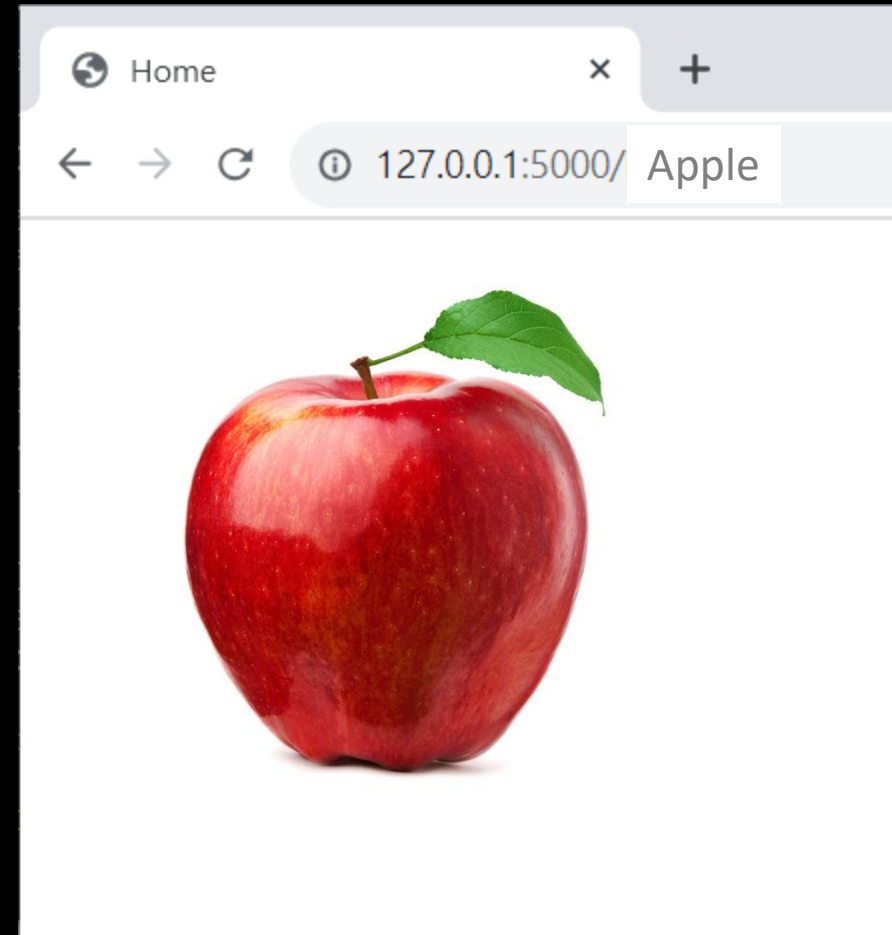
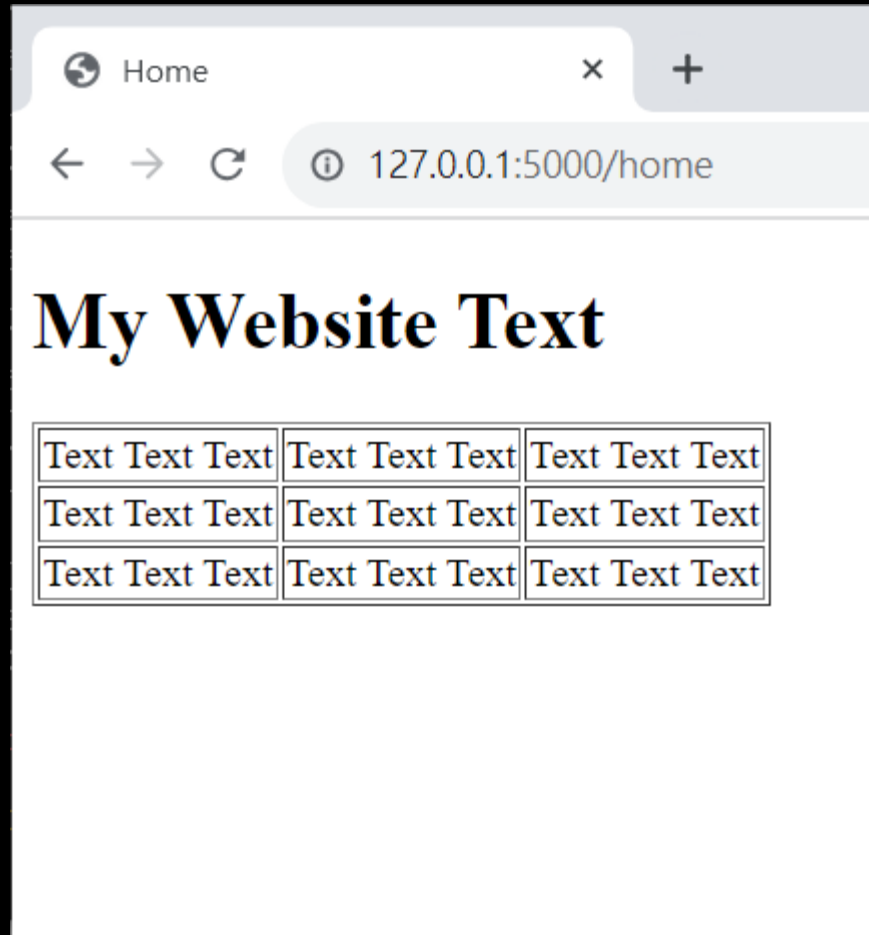
# Exercise 45: URL Info

- Show the information in the URL.
- For example:



# Exercise 46: Flask Load HTML

- Load different HTML for different URL
- For example:





# Exercise 47: Show Variables

- There is a Python object

```
x = {  
    "name": "John",  
    "age": 30,  
    "city": "New York"  
}
```

Show the object variables on HTML

# Exercise 48: Show double of the inputted number

## double.py

```
from flask import Flask, request, render_template

app = Flask(__name__)

# Display input page
@app.route("/")
def index():
    return render_template("index.html")

# process the submission
@app.route("/predict", methods=["POST"])
def predict():
    x = int(request.form["x"])    # Take data from form
    result = x * 2
    return render_template("index.html", result=result) #
update the result

if __name__ == "__main__":
    app.run(debug=True)
```

## Index.html

```
<!DOCTYPE html>
<html>
<head>
    <title>Simple Flask Form (Compute the
double)</title>
</head>
<body>

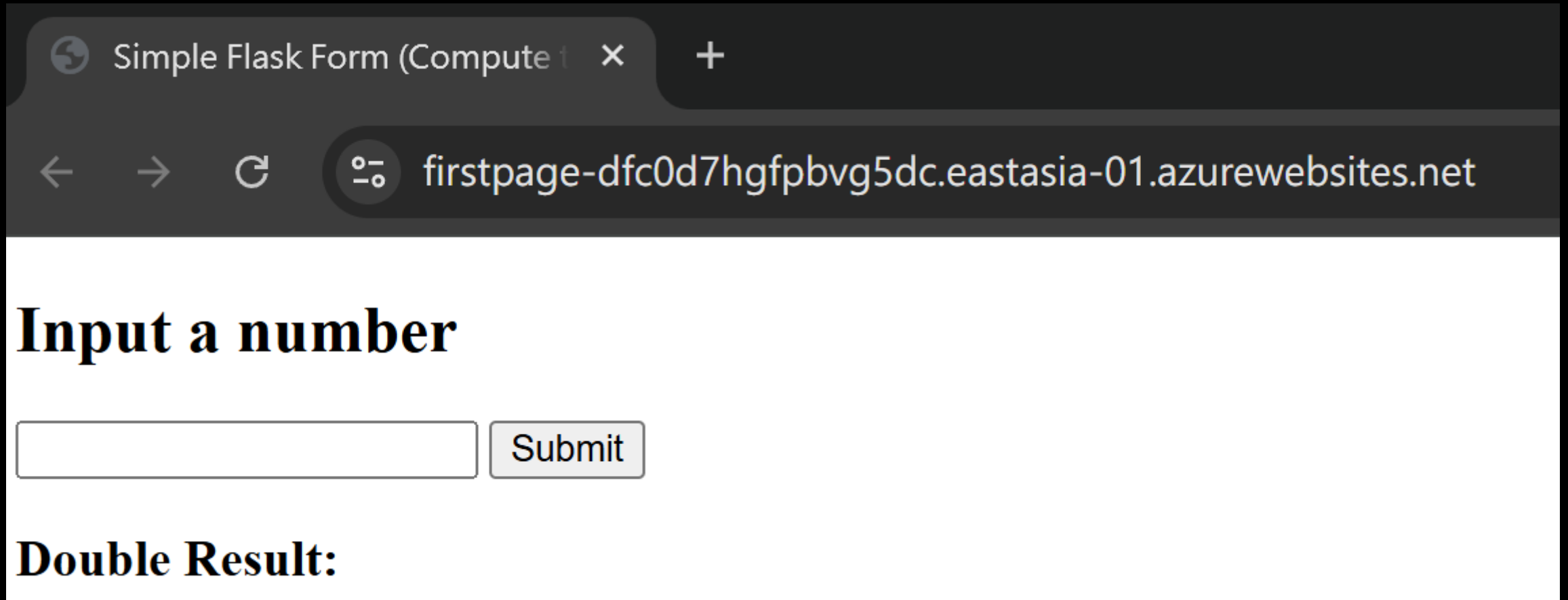
    <h2>Input a number</h2>

    <!-- form send to /predict -->
    <form action="/predict" method="post">
        <input type="number" name="x" required>
        <button type="submit">Submit</button>
    </form>

    <!-- display the result -->
    {% if result is not none %}
        <h3>Double Result: {{ result }}</h3>
    {% endif %}

</body>
</html>
```

# Exercise 48: Show double of the inputted number (cont.)



The screenshot shows a web browser window with a single tab titled "Simple Flask Form (Compute...". The address bar displays the URL "firstpage-dfc0d7hgfpbvg5dc.eastasia-01.azurewebsites.net". The page content includes a heading "Input a number" in a large, bold, black serif font. Below the heading is a text input field and a "Submit" button. Underneath the input field is the text "Double Result:" in a bold, black serif font.

Simple Flask Form (Compute... x +

← → ↻ 🏠 firstpage-dfc0d7hgfpbvg5dc.eastasia-01.azurewebsites.net

**Input a number**

Submit

**Double Result:**

- Give me the link!